

RAI 818 – INTELLIGENT MOBILE **ROBOTICS**

EKF SLAM

- **Problem Definition**
- **Algorithm**
- **Implementation**

Text : Principles of Robot Motion – Theory, Algorithms and Implementations
H. Choset, K.M. Lynch, S. Hutchinson, G. Kantor, W. Burgard, et al.

INSTRUCTOR:

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EKF SLAM

• Definition of the Slam Problem

Definition of the SLAM Problem

• Given

- The robot's controls $u_{1:T} = \{u_1, u_2, u_3, \dots, u_T\}$
- Observations $z_{1:T} = \{z_1, z_2, z_3, \dots, z_T\}$

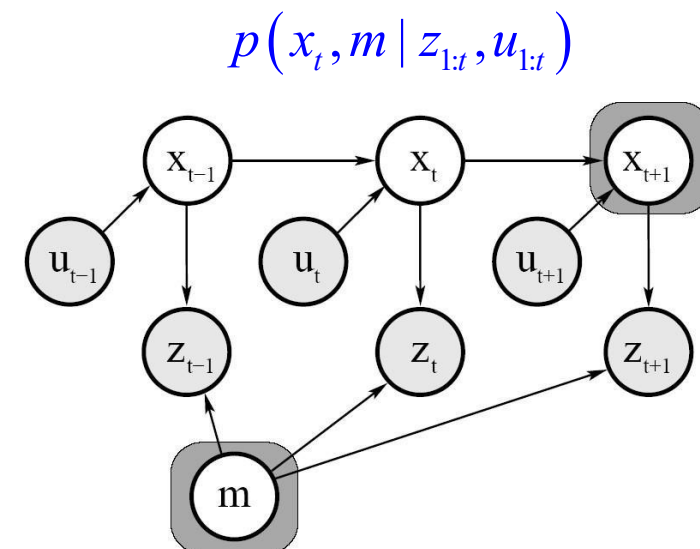
• Wanted

- Map of the environment m
- Path of the robot $x_{0:T} = \{x_0, x_1, x_2, \dots, x_T\}$

• Bayes Filter

- Recursive filter with prediction and correction step
- Estimates $p(x_t, m | z_{1:t}, u_{1:t})$
- Kalman Filter is a recursive Bayes filter for the linear Gaussian case
- EKF for dealing with non-linearities

- We consider here the Kalman filter as a solution to the online SLAM problem



EKF SLAM

- Definition of the Slam Problem
- EKF SLAM Algorithm

Algorithm

1: **Extended Kalman Filter (EKF)**
 $(\vec{x}_{t-1}, P_{t-1}, u_t, z_t)$

2: $\vec{x}_t = f(\vec{x}_{t-1}, u_t)$

3: $\vec{P}_t = F_t P_{t-1} F_t^T + Q_t$

4: $K_t = \vec{P}_t H_t^T (H_t \vec{P}_t H_t^T + R)^{-1}$

5: $x_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$

6: $P_t = \vec{P}_t - K_t H_t \vec{P}_t$



EKF SLAM

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Algorithm

- Application of the EKF to SLAM
- Estimate the robot's pose and locations of landmarks in the environment
- Assumption: known correspondences
- State space (for the 2D plane) is
 - Map with n landmarks: $(3+2n)$ – dimensional Gaussian
 - Belief is represented by

$$x_t = \left(\underbrace{x, y, \theta}_{\text{robot's pose}} \quad \underbrace{m_{1,x}, m_{1,y}, \dots, m_{n,x}, m_{n,y}}_{\text{landmark } 1 \quad \text{landmark } n} \right)$$

$$\underbrace{\begin{pmatrix} x \\ y \\ \theta \\ m_{1,x} \\ m_{1,y} \\ \vdots \\ m_{n,x} \\ m_{n,y} \end{pmatrix}}_x \underbrace{\begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{x\theta} & \sigma_{xm_{1,x}} & \sigma_{xm_{1,y}} & \dots & \sigma_{xm_{n,x}} & \sigma_{xm_{n,y}} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{y\theta} & \sigma_{ym_{1,x}} & \sigma_{ym_{1,y}} & \dots & \sigma_{ym_{n,x}} & \sigma_{ym_{n,y}} \\ \sigma_{\theta x} & \sigma_{\theta y} & \sigma_{\theta\theta} & \sigma_{\theta m_{1,x}} & \sigma_{\theta m_{1,y}} & \dots & \sigma_{\theta m_{n,x}} & \sigma_{\theta m_{n,y}} \\ \sigma_{m_{1,x}x} & \sigma_{m_{1,x}y} & \sigma_{m_{1,x}\theta} & \sigma_{m_{1,x}m_{1,x}} & \sigma_{m_{1,x}m_{1,y}} & \dots & \sigma_{m_{1,x}m_{n,x}} & \sigma_{m_{1,x}m_{n,y}} \\ \sigma_{m_{1,y}x} & \sigma_{m_{1,y}y} & \sigma_{m_{1,y}\theta} & \sigma_{m_{1,y}m_{1,x}} & \sigma_{m_{1,y}m_{1,y}} & \dots & \sigma_{m_{1,y}m_{n,x}} & \sigma_{m_{1,y}m_{n,y}} \\ \vdots & \vdots & \vdots & \vdots & \vdots & \ddots & \vdots & \vdots \\ \sigma_{m_{n,x}x} & \sigma_{m_{n,x}y} & \sigma_{m_{n,x}\theta} & \sigma_{m_{n,x}m_{1,x}} & \sigma_{m_{n,x}m_{1,y}} & \dots & \sigma_{m_{n,x}m_{n,x}} & \sigma_{m_{n,x}m_{n,y}} \\ \sigma_{m_{n,y}x} & \sigma_{m_{n,y}y} & \sigma_{m_{n,y}\theta} & \sigma_{m_{n,y}m_{1,x}} & \sigma_{m_{n,y}m_{1,y}} & \dots & \sigma_{m_{n,y}m_{n,x}} & \sigma_{m_{n,y}m_{n,y}} \end{pmatrix}}_P$$



EKF SLAM

- Definition of the Slam Problem
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Algorithm

- More Compactly

$$\underbrace{\begin{pmatrix} x_R \\ m_1 \\ \vdots \\ m_n \end{pmatrix}}_x \quad \underbrace{\begin{pmatrix} \Sigma_{x_R x_R} & \Sigma_{x_R m_1} & \dots & \Sigma_{x_R m_n} \\ \Sigma_{m_1 x_R} & \Sigma_{m_1 m_1} & \dots & \Sigma_{m_1 m_n} \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma_{m_n x_R} & \Sigma_{m_n m_1} & \dots & \Sigma_{m_n m_n} \end{pmatrix}}_P$$

- Even More Compactly

$$\underbrace{\begin{pmatrix} x \\ m \end{pmatrix}}_x \quad \underbrace{\begin{pmatrix} \Sigma_{xx} & \Sigma_{xm} \\ \Sigma_{mx} & \Sigma_{mm} \end{pmatrix}}_P$$



EKF SLAM



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- Definition of the Slam Problem
- EKF SLAM Algorithm
 - Filter Cycle

EKF SLAM: Filter Cycle

1. State prediction
2. Measurement prediction
3. Measurement
4. Data association
5. Update

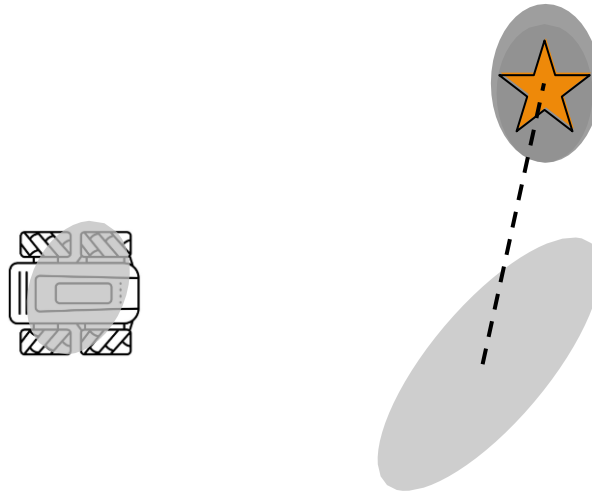


Defining futures

EKF SLAM

- Definition of the Slam Problem
- EKF SLAM Algorithm
 - Filter Cycle
 - **State Prediction**

State Prediction



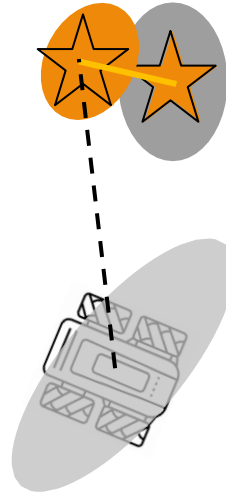
$$\underbrace{\begin{pmatrix} x_R \\ m_1 \\ \vdots \\ m_n \end{pmatrix}}_x \quad \underbrace{\begin{pmatrix} \Sigma x_R x_R & \Sigma x_R m_1 & \dots & \Sigma x_R m_n \\ \Sigma m_1 x_R & \Sigma m_1 m_1 & \dots & \Sigma m_1 m_n \\ \vdots & \vdots & \ddots & \vdots \\ \Sigma m_n x_R & \Sigma m_n m_1 & \dots & \Sigma m_n m_n \end{pmatrix}}_P$$



EKF SLAM

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 - Filter Cycle
 - **State Prediction**

State Prediction



$$\underbrace{\begin{pmatrix} x_R \\ m_1 \\ \vdots \\ m_n \end{pmatrix}}_x \quad \underbrace{\begin{pmatrix} \sum x_R x_R & \sum x_R m_1 & \dots & \sum x_R m_n \\ \sum m_1 x_R & \sum m_1 m_1 & \dots & \sum m_1 m_n \\ \vdots & \vdots & \ddots & \vdots \\ \sum m_n x_R & \sum m_n m_1 & \dots & \sum m_n m_n \end{pmatrix}}_P$$



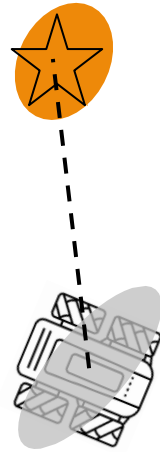
EKF SLAM



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State Prediction



$$\underbrace{\begin{pmatrix} x_R \\ m_1 \\ \vdots \\ m_n \end{pmatrix}}_x \quad \underbrace{\begin{pmatrix} \sum x_R x_R & \sum x_R m_1 & \dots & \sum x_R m_n \\ \sum m_1 x_R & \sum m_1 m_1 & \dots & \sum m_1 m_n \\ \vdots & \vdots & \ddots & \vdots \\ \sum m_n x_R & \sum m_n m_1 & \dots & \sum m_n m_n \end{pmatrix}}_P$$



Defining futures

EKF SLAM

- Definition of the Slam Problem
- **EKF SLAM Algorithm**
 - Filter Cycle
 - State Prediction
 - **Setup**

Setup

- Robot moves in the 2D plane
- Velocity-based motion model
- Robot observes point landmarks
- Range-bearing sensor
- Known data association
- Known number of landmarks



Defining futures

EKF SLAM

- Definition of the Slam Problem
- **EKF SLAM Algorithm**
 - Filter Cycle
 - State Prediction
 - Setup
 - **Initialization**

Initialization

- Robot starts in its own reference frame (all landmarks unknown)
- $2N+3$ dimensions

$$x_0 = (0 \quad 0 \quad 0 \quad \dots \quad 0)^T$$

$$P_0 = \begin{pmatrix} 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & 0 & \dots & 0 \\ 0 & 0 & 0 & \infty & \dots & 0 \\ \vdots & \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & 0 & \dots & \infty \end{pmatrix}$$



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3: $\vec{P}_t = F_t P_{t-1} F_t^T + Q_t$

4: $K_t = \vec{P}_t H_t^T (H_t \vec{P}_t H_t^T + R)^{-1}$

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EKF SLAM

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 - **Prediction Step**
 - Motion Update

Motion Update

- Goal: Update state space based on the robot's motion
- Robot motion in the plane

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \underbrace{\begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}}_{f_{x,y,\theta}(u_t, (x, y, \theta)^T)}$$

- How to map that to the $2N + 3$ dim space?



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 - State Space Update

State Space Update

- From the motion in the plane

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

- To the $2N + 3$ dimensional space?

$$\begin{pmatrix} x' \\ y' \\ \theta' \end{pmatrix} = \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \underbrace{\begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \end{pmatrix}^T}_{\underbrace{G_x^T}_{2N \text{ cols}}} \underbrace{\begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}}_{f(u_t, x_t)}$$



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 - State Space Update
 - **Covariance Update**

Covariance Update

- The function F only affects the robot's motion and not the landmarks

Jacobian of the motion (3×3)

$$F_t = \begin{pmatrix} F_t^x & 0 \\ 0 & I \end{pmatrix} \quad (2N \times 2N)$$

$$F_t^x = \frac{\partial}{\partial (x, y, \theta)^T} \begin{pmatrix} x \\ y \\ \theta \end{pmatrix} + \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix} = I + \frac{\partial}{\partial (x, y, \theta)^T} \begin{pmatrix} -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos \theta - \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$F_t^x = \begin{bmatrix} 1 & 0 & -\frac{v_t}{\omega_t} \cos \theta + \frac{v_t}{\omega_t} \cos(\theta + \omega_t \Delta t) \\ 0 & 1 & -\frac{v_t}{\omega_t} \sin \theta + \frac{v_t}{\omega_t} \sin(\theta + \omega_t \Delta t) \\ 0 & 0 & 1 \end{bmatrix}$$



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5: $x_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$

6: $P_t = \vec{P}_t - K_t H_t \vec{P}_t$

$$\begin{aligned} \vec{P}_t &= F_t P_{t-1} F_t^T + Q_t \\ &= \begin{pmatrix} F_t^x & 0 \\ 0 & I \end{pmatrix} \begin{pmatrix} P_{xx} & P_{xm} \\ P_{mx} & P_{mm} \end{pmatrix} \begin{pmatrix} (F_t^x)^T & 0 \\ 0 & I \end{pmatrix} + Q_t \\ &= \begin{pmatrix} F_t^x P_{xx} (F_t^x)^T & F_t^x P_{xm} \\ (F_t^x)^T P_{mx} & P_{mm} \end{pmatrix} + Q_t \end{aligned}$$



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 - Motion Update
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 - Covariance Update
 - **Prediction Pseudo Code**

Complete Prediction step

$$2: G_x = \begin{pmatrix} 1 & 0 & 0 & 0 & \dots & 0 \\ 0 & 1 & 0 & 0 & \dots & 0 \\ 0 & 0 & 1 & 0 & \dots & 0 \end{pmatrix}$$

$$5: \bar{P}_t = F_t P_{t-1} F_t^T + G_x^T Q_t^x G_x$$

$$3: \bar{x}_t = x_{t-1} + G_x^T \begin{pmatrix} -\frac{v_t}{\omega_t} \sin x_{t-1}, \theta + \frac{v_t}{\omega_t} \sin(x_{t-1}, \theta + \omega_t \Delta t) \\ \frac{v_t}{\omega_t} \cos x_{t-1}, \theta - \frac{v_t}{\omega_t} \cos(x_{t-1}, \theta + \omega_t \Delta t) \\ \omega_t \Delta t \end{pmatrix}$$

$$4: F_t = I + G_x^T \begin{pmatrix} 0 & 0 & -\frac{v_t}{\omega_t} \cos x_{t-1}, \theta + \frac{v_t}{\omega_t} \cos(x_{t-1}, \theta + \omega_t \Delta t) \\ 0 & 0 & -\frac{v_t}{\omega_t} \sin x_{t-1}, \theta + \frac{v_t}{\omega_t} \sin(x_{t-1}, \theta + \omega_t \Delta t) \\ 0 & 0 & 0 \end{pmatrix}$$



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3: ~~$\vec{P}_t = F_t P_{t-1} F_t^T + Q_t$~~ **Done**

4: $K_t = \vec{P}_t H_t^T (H_t \vec{P}_t H_t^T + R)^{-1}$

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 - **Correction Step**

Correction Step

- Known data association
- $c_i^t = j$: i -th measurement at time t observes the landmark with index j
- Initialize landmark if unobserved
- Compute the expected observation
- Compute the Jacobian of h
- Proceed with computing the Kalman gain
- Range observation $z_t^i = (r_t^i, \phi_t^i)^T$
- If landmark has not been observed

$$\begin{pmatrix} \vec{x}_{j,x} \\ \vec{x}_{j,y} \end{pmatrix} = \begin{pmatrix} \vec{x}_{t,x} \\ \vec{x}_{t,y} \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i) + \vec{x}_{t,\theta} \\ r_t^i \sin(\phi_t^i) + \vec{x}_{t,\theta} \end{pmatrix}$$

observed location of
landmark, j

estimated robot
location

relative
measurement

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 - **Expected Observation**

Expected Observation

- Compute expected observation according to the current estimate

$$\delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} + \begin{pmatrix} \bar{x}_{j,x} - \bar{x}_{t,x} \\ \bar{x}_{j,y} - \bar{x}_{t,y} \end{pmatrix}$$

$$q = \delta^T \delta$$

$$\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ a \tan 2(\delta_y, \delta_x) - \bar{\mu}_{t,\theta} \end{pmatrix}$$

$$= h(\bar{x}_t)$$

- Compute the Jacobian

$$\begin{matrix} \uparrow \\ \text{low} \end{matrix} H_t^i = \frac{\partial h(\bar{x}_t)}{\partial \bar{x}_t} = \begin{pmatrix} \frac{\partial \sqrt{q}}{\partial x} & \frac{\partial \sqrt{q}}{\partial y} & \dots \\ \frac{\partial a \tan 2(\dots)}{\partial x} & \frac{\partial a \tan 2(\dots)}{\partial y} & \dots \end{pmatrix} \quad \begin{aligned} \frac{\partial \sqrt{q}}{\partial x} &= \frac{1}{2} \frac{1}{\sqrt{q}} 2\delta_x (-1) \\ &= \frac{1}{q} (-\sqrt{q} \delta_x) \end{aligned}$$

Low dimensional space (x, y, θ, m_{j,x}, m_{j,y})



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$$q = \delta^T \delta$$

$$\hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ a \tan 2(\delta_y, \delta_x) - \bar{\mu}_{t,\theta} \end{pmatrix}$$

$$= h(\bar{x}_t)$$

- Compute the Jacobian

$$\overset{\text{low}}{H}_t^i = \frac{\partial h(\bar{x}_t)}{\partial \bar{x}_t} = \frac{1}{q} \begin{pmatrix} -\sqrt{q} \partial x & -\sqrt{q} \partial y & 0 & \sqrt{q} \partial x & \sqrt{q} \partial y \\ \partial y & -\partial x & -q & -\partial y & \partial x \end{pmatrix}$$

Low dimensional space (x, y, θ, m_{j,x}, m_{j,x})

Map to high dimensional space

$$\overset{\text{low}}{H}_i^j \xrightarrow{F_{x,j}} F_{x,j} = \begin{pmatrix} 1 & 0 & 0 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 1 & 0 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 1 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & 0 \dots 0 & 1 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & \underbrace{0 \dots 0}_{2j-2} & 0 & 1 & \underbrace{0 \dots 0}_{2N-2j} \end{pmatrix}$$



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2: ~~$\vec{x}_t = f(\vec{x}_{t-1}, u_t)$~~ **Done**

3: ~~$\vec{P}_t = F_t P_{t-1} F_t^T + Q_t$~~ **Done**

4: ~~$K_t = \vec{P}_t H_t^T (H_t \vec{P}_t H_t^T + R)^{-1}$~~ **Done**

5: ~~$\vec{x}_t = \vec{x}_t + K_t (z_t - h(\vec{x}_t))$~~ **Done**

6: ~~$P_t = \vec{P}_t - K_t H_t \vec{P}_t$~~ **Done**



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 - **Correction Pseudo Code**

Complete Correction Step

$$6: R_t = \begin{pmatrix} \sigma_r^2 & 0 \\ 0 & \sigma_\phi^2 \end{pmatrix}$$

7: For all observed features $z_t^i = (r_t^i, \phi_t^i)^T$ do

$$8: j = c_i^T$$

9: if landmark j never seen before

$$10: \begin{pmatrix} \bar{x}_{j,x} \\ \bar{x}_{j,y} \end{pmatrix} = \begin{pmatrix} \bar{x}_{t,x} \\ \bar{x}_{t,y} \end{pmatrix} + \begin{pmatrix} r_t^i \cos(\phi_t^i + \bar{x}_{t,\theta}) \\ r_t^i \sin(\phi_t^i + \bar{x}_{t,\theta}) \end{pmatrix}$$

11: end if

$$12: \delta = \begin{pmatrix} \delta_x \\ \delta_y \end{pmatrix} + \begin{pmatrix} \bar{x}_{j,x} - \bar{x}_{t,x} \\ \bar{x}_{j,y} - \bar{x}_{t,y} \end{pmatrix}$$

$$13: q = \delta^T \delta$$

$$14: \hat{z}_t^i = \begin{pmatrix} \sqrt{q} \\ a \tan 2(\delta_y, \delta_x) - \bar{x}_{t,\theta} \end{pmatrix} \\ = h(\bar{x}_t)$$



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Complete Correction Step

- 15:
$$F_{x,j} = \begin{pmatrix} 1 & 0 & 0 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 1 & 0 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 1 & 0 \dots 0 & 0 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & 0 \dots 0 & 1 & 0 & 0 \dots 0 \\ 0 & 0 & 0 & \underbrace{0 \dots 0}_{2j-2} & 0 & 1 & \underbrace{0 \dots 0}_{2N-2j} \end{pmatrix}$$
- 16:
$$H_i^T = \frac{1}{q} \begin{pmatrix} -\sqrt{q} \partial x & -\sqrt{q} \partial y & 0 & \sqrt{q} \partial x & \sqrt{q} \partial y \\ \partial y & -\partial x & -q & -\partial y & \partial x \end{pmatrix} F_{x,j}$$
- 17:
$$K_t = P_t H_t^T (H_t P_t H_t^T + R_t)^{-1}$$
- 18:
$$x_t = \bar{x}_t + K_t (z_t - h(\bar{x}_t))$$
- 19:
$$P_t = \bar{P}_t - K_t H_t \bar{P}_t$$
- 20: *end For*
- 21:
$$x_t = \bar{x}_t$$
- 22:
$$P_t = \bar{P}_t$$
- 23: *return* x_t, P_t



EKF SLAM

- Definition of the Slam Problem
- **EKF SLAM Algorithm**
 - Filter Cycle
 - State Prediction
 - Setup
 - Initialization
 - Prediction Step
 - Correction Step
 - **Implementation Notes**

Implementation Notes

- Measurement update in a single step requires only one full belief update
- Always normalize the angular components
- You may not need to create the G matrices explicitly (e.g., in Matlab)
- DONE!



Defining futures

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Loop Closing

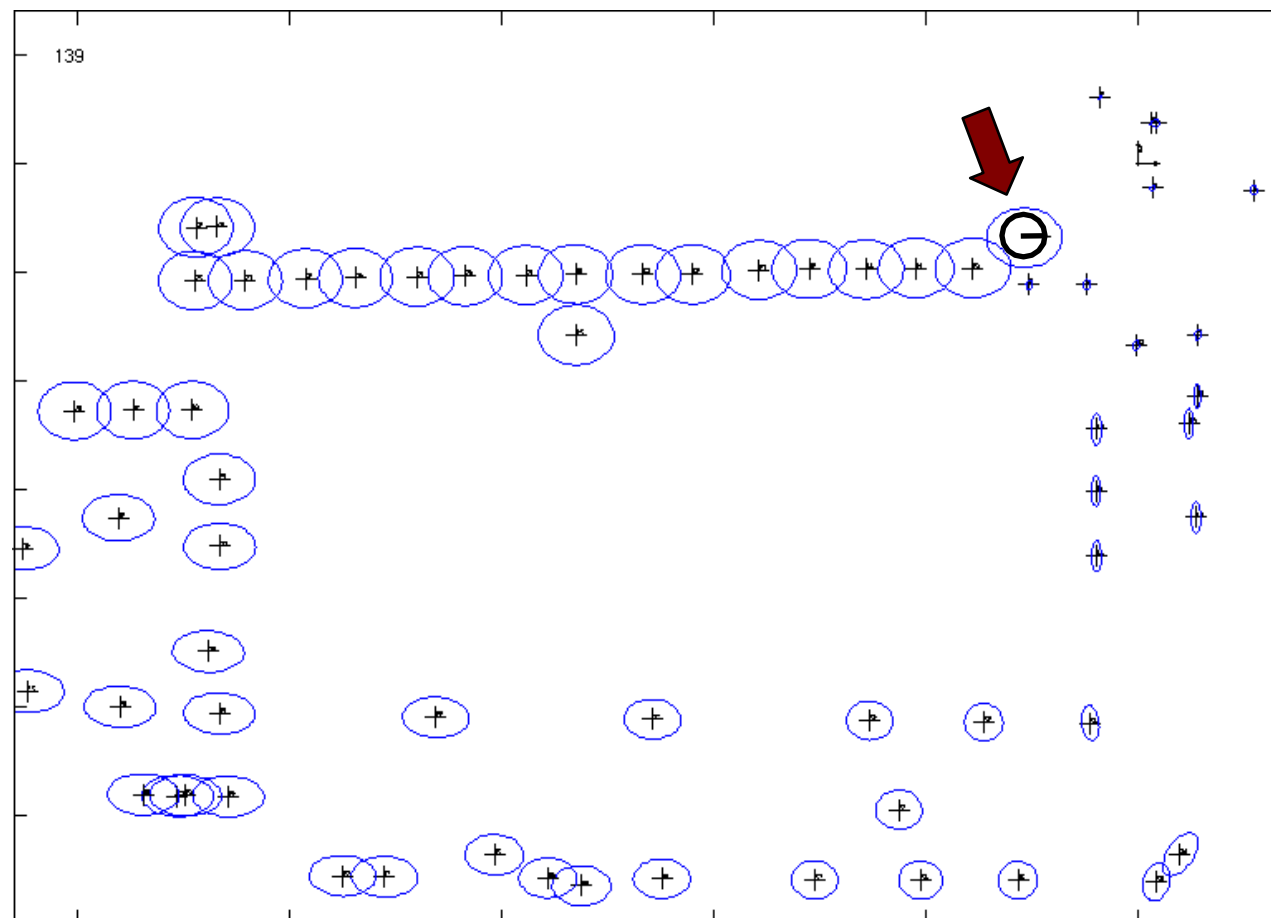
- *Loop closing means recognizing an already mapped area*
- *Data association under*
 - *high ambiguity*
 - *possible environment symmetries*
- *Uncertainties collapse after a loop closure (whether the closure was correct or not)*



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Loop Closing – Before Loop Closure

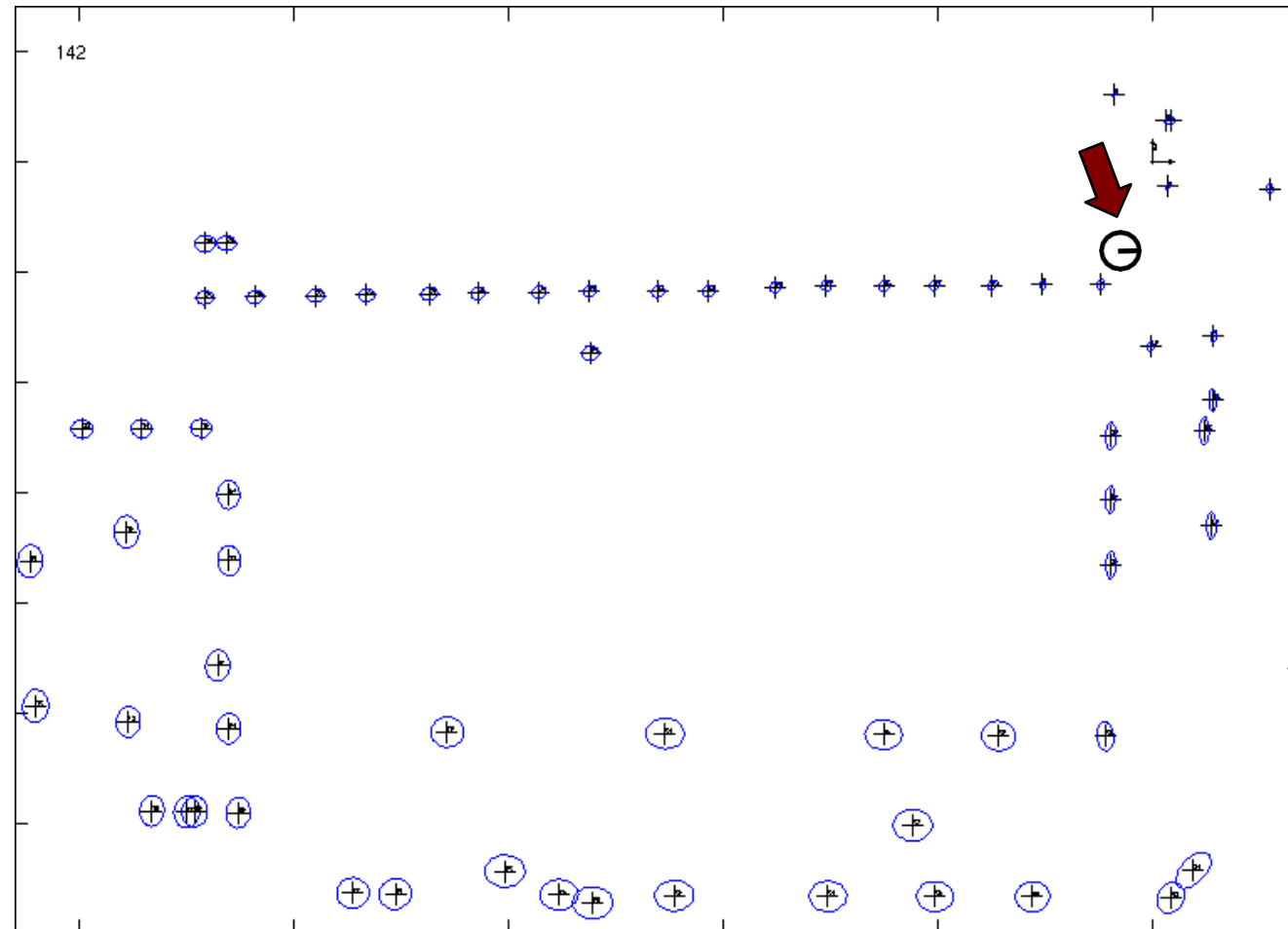


Defining futures

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Loop Closing in SLAM

- Loop closing **reduces** the uncertainty in robot and landmark estimates
- This can be exploited when exploring an environment for the sake of better (e.g. more accurate) maps
- Wrong loop closures lead to filter divergence



Defining futures